**DO NOW — RECALL**

# Lesson 084 · A* Search: g, h and f Scores

Name: _____ Class: _____ Date: _____

**1. What is Dijkstra's tentative distance?**

**2. What does f(n) equal in A*?**

**3. What is an admissible heuristic?**

**4. Binary search target12 in sorted [2,4,6,8,10,12,14]. Show midpoint indexes.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Known best route cost from source.

### 2. Model answer

g(n)+h(n).

### 3. Model answer

Never overestimates true remaining cheapest cost.

### 4. Model answer

mid3=8 then mid5=12; return index5.

### By the end, you can…

• A* orders candidate states using f(n)=g(n)+h(n), where g is known cost from the start and h estimates remaining cost to the goal.

• A heuristic that never overestimates remaining true cost is admissible; with suitable search implementation/assumptions this supports optimality.

• Track open candidates, their g/h/f and predecessors. An overestimating heuristic or incorrect update can make search miss the shortest path.

### Vocabulary — know these terms

A* — Heuristic-guided shortest-path search.

g(n) — Cost so far from start.

h(n) — Estimated cost to goal.

f(n) — g(n)+h(n) priority.

NEW LEARNING

# Read, annotate and explain

### Observe

A* orders candidate states using f(n)=g(n)+h(n), where g is known cost from the start and h estimates remaining cost to the goal.

### Apply

A heuristic that never overestimates remaining true cost is admissible; with suitable search implementation/assumptions this supports optimality.

### Evaluate

Track open candidates, their g/h/f and predecessors. An overestimating heuristic or incorrect update can make search miss the shortest path.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**Find P→R: P→Q = 3, P→R = 8, Q→R = 2. Heuristics: h(P)=5, h(Q)=2, h(R)=0.**

### Calculate all frontier f-scores

From P: Q has g=3,h=2,f=5 ; R has g=8,h=0,f=8 . Put both in OPEN. Choose Q next (smallest f), not simply the nearest-looking node.

### Improve the path to R

Via Q→R, new g(R)=3+2=5 , which improves the existing 8. Thus f(R)=5+0=5 , and predecessor R becomes Q. OPEN now contains R with the improved values.

### Confirm the path and heuristic meaning

Select goal R from OPEN, then reconstruct R←Q←P to obtain P→Q→R , cost 5 . The supplied heuristics do not overestimate remaining true cost; the next lesson can explore what happens when this fails.

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

After expanding A, calculate f for B and C using stated graph/heuristics. [6]

### Question 2

After expanding B then C, show improvements to g(C) and g(D); state best path. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Compare A* with Dijkstra when h=0 everywhere, and explain risk of overestimation. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A* h(n) is the exact distance to the goal for every possible heuristic.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.